HT Group	Name

CATHOLIC JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS

H1 BIOLOGY **8875/02**

Paper 2 Core 27 August 2008 Wednesday

2 hours

Additional materials: Writing Paper

Section 1.01

READ THESE INSTRUCTIONS FIRST

Write your name and HT group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section 1.02 Section A

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section 1.03 Section B

Answer **one** question.

Write your answers on the separate answer paper.

All working for numerical answers must be shown.

At the end of the examination,

1. fasten all your work securely together;
2. enter the number of the Section B question you have answered in the grid opposite.

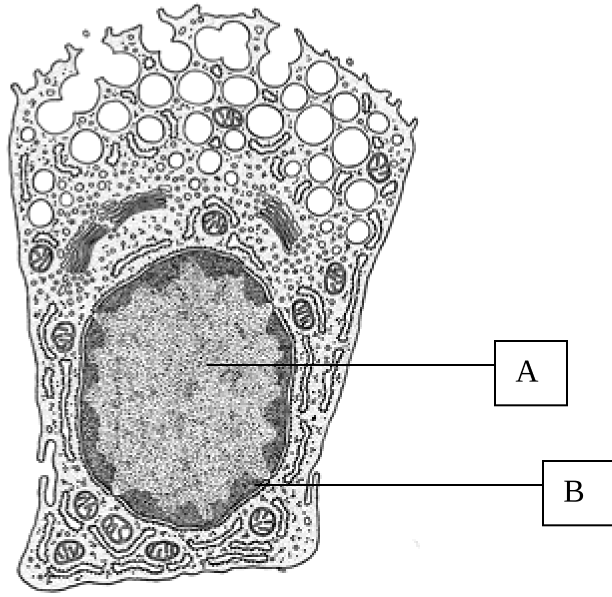
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A	
1 [10]	
2 [13]	
3 [12]	
4 [15]	
5 [17]	
6 [13]	
Section B	
Total	

This document consists of **11** printed pages.

1 The diagram below shows a secretory cell from the liver.

ANSWERS



1 (a) State the names of part A and B [2]

A – Euchromatin.
B – Heterochromatin.

(b) Explain how the staining pattern in the nucleus are related to the cell's activity. [2]

1 **Nucleus with more light – stained areas than dark; has fewer** heterochromatin clumps;
2 **more DNA surface** available; **for transcription** of genetic information;
3 **Cell activity is dependent on level of coiling of DNA into chromatin.**

Some cells similar to that shown in the diagram were grown in a culture. Radioactive amino acids were added to the solution in which they were being grown. The radioactivity acts as a label on the amino acid so that it can be detected wherever it is. This radioactive label allows amino acids to be followed through the cell.
At various times, samples of the cells were taken and the amount of radioactivity in different organelles was measured. The results are shown in the table.

Time after radioactive amino acids were added to the solution/ minutes	Amount of radioactivity present/arbitrary units		
	Golgi apparatus	Rough endoplasmic reticulum	Vesicles
1	21	120	6
20	42	68	6
40	86	39	8
60	76	28	15
90	50	27	28
120	38	26	56

(c) What is the fate of amino acids that reach the endoplasmic reticulum? [2]

1 carried to **mRNA that is bound to ribosome**; and **peptide bond** is formed between 2 amino acids;
2 type of reaction: **Condensation to form polypeptides**
3 **Polypeptides →** fold to become **functional proteins** in rough ER and Golgi
4 **packaged in vesicles** and **sent to Golgi** apparatus for modification;

(d) Account for the difference in radioactivity observed in the organelles after 60 minutes. [4]

1 Rough ER: **amino acids are joined by peptide bond to form polypeptide** in rough ER; amino acids are readily **incorporated to the growing chain of polypeptide;**
2 Golgi apparatus**: raw polypeptides are processed via modification in GA**; carbohydrate chains may be added to polypeptides transported there from rough **ER to become glycoproteins;**
3 Vesicles: **Vesicles carrying finished glycoproteins** or processed proteins are budded off from GA;
4 **Comparatively least amount of radioactivity observed** here due to the **secretion of proteins** out of cell; accounts for the fall of radioactivity;

[Total: 10]

2 Two closely related species of frog, *Hyla ewingi* and *Hyla verrauxi* live in South Australia.

(a) Explain what is meant by the terms;

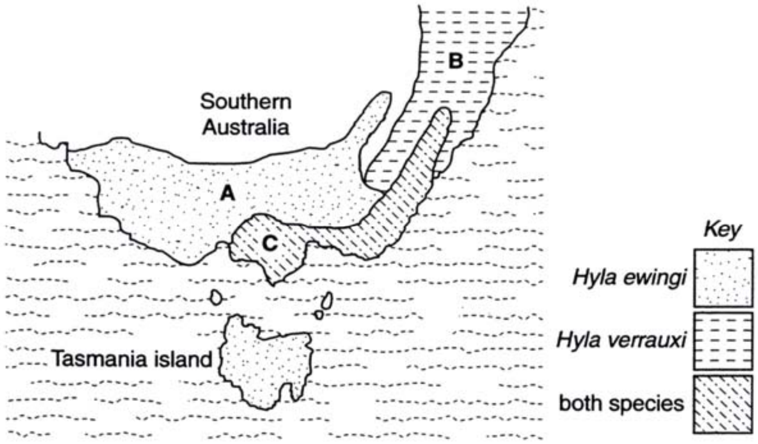
(i) *Niche* [2]

1 Niche is the **unique position of an organism in an ecosystem**
2 It include **descriptions of the organism's life history, habitat and food chain**
3 It describes **how a population responds** to the **abundance of its resources and enemies**
4 Niche also describes **how the various abiotic factors influences** how **populations affect,and are affected by resources and enemies**
5 **No 2 species** can **occupy in the same niche** for a long time.

(ii) Species [3]

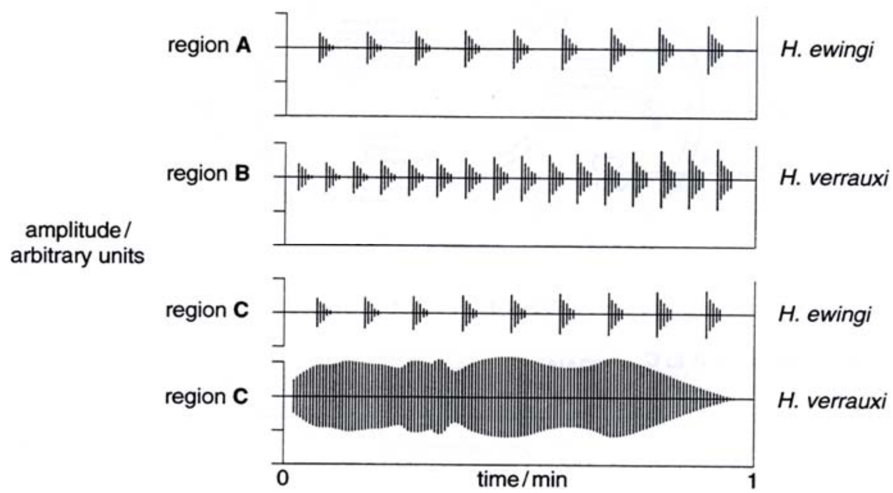
1 Similar **morphological, physiological, biochemical** and behavioural features;
2 **DNA** of organisms of a **species nearly identical/have identical gene loci.**
3 Interbreed/ **reproduce to produce fertile offspring;**
4 **Reproductively isolated**;
5 Occupy **same (ecological) niche**;

The figure below shows the distribution of the tree frogs in Southern Australia.



Hyla ewingi and *Hyla verrauxi* are two closely related species of tree frogs from southern Australia. Research from breeding studies and DNA sequence data has shown that they have weak genetic incompatibility.

Male frogs attract females of the same species for mating by their pulsing call. The pulse rate of the male calls of the two species is almost identical. However, when both species coexist within the same region, the calls of *H. ewingi* are quite different than those of *H. verrauxi*.
The following data shows the calls of *H. ewingi* and *H. verrauxi* from three regions, A, B and C. The male advertisement calls were recorded and played back electronically to produce the pattern shown



Some female frogs of the species *H. verrauxi* were transferred from region B to region C. Observations shown that they were not attracted by the calls of the males in region C.

(b) Suggest why this is so. [3]

1 The **male advertisement calls** by *H. verrauxi* in region C are **no longer pulsatile / discontinous**
2 **More continuous** without **any interval in C;**
3 **Amplitude** of the sound waves also differ: Sound intensity of male call in region B **is from soft to loud** compared to **loud to soft in C.**

(c) Explain how the process of natural selection could have led to the different male advertisement call in *H. verrauxi* in region C. [5]

1 **Variation** in male advertisement calls present **in population of Hyla frogs in region C**;
2 As *H. ewingi* and *H. verrauxi* have **similar advertisement calls**; **competition for mates** exists between these two species;
3 **Competition for mates** (sexual selection by females) in the same habitat **selects males with a different/ continuous calls** compared to **males with a pulsatile call;**
4 **Successful male** mates with **female *Hyla verrauxi* tree frogs**;
5 **reproduced more offspring** with **similar male advertisement call**;
6 As these **genes that** code for such **male advertisement** calls are passed down to the next generation;
7 Those *H. verrauxi* males **who fail to mate** with another female will **diminish in numbers** after several generations;
8 Over time, **region C sees an evolution** of a **different male advertisement call** for *H. verrauxi*

[Total: 13]

3 Fig. 3.1 and 3.2 are diagrams showing transcription and translation.

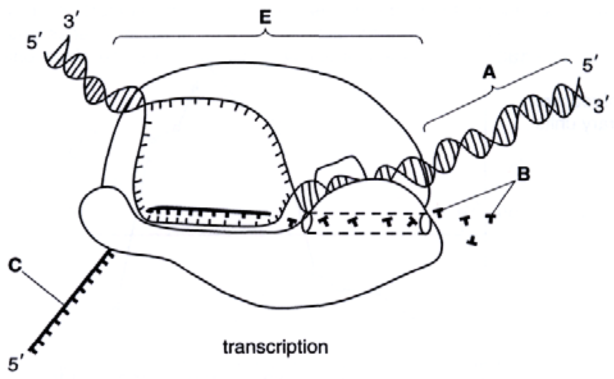


Fig. 3.1

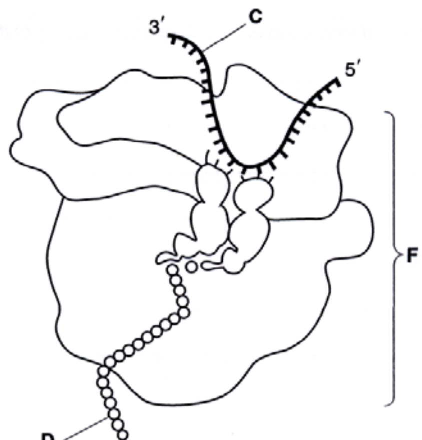


Fig. 3.2

(a) Identify structures A to D and describe their function. [4]

A: DNA (deoxyribonucleic acid)
B: ribonucleotides
C: messenger RNA
D: polypeptide (R) protein

(b) State where in the cell does transcription and translation occurs. [2]
Transcription: Nucleus
Translation: Cytoplasm.

(c) List four additional ways in which transcription differs from translation. [4]

1 **activated transfer RNA** carries amino acids used in translation; **no such carriers needed for transcription;**
2 **ribonucleotides** are building blocks in transcription; **amino acids** building blocks in translation;
3 **mRNA is the template** for translation; **DNA is the template** for transcription;
4 ribonucleotides are linked by **phosphodiester bonds**; amino acids are linked by **peptide bonds;**
5 **RNA polymerase begins transcription from 3’ to 5’ on antisense strand**, **elongation of mRNA from 5’ to 3’**; **ribosome translocates from 5’ to 3’;**

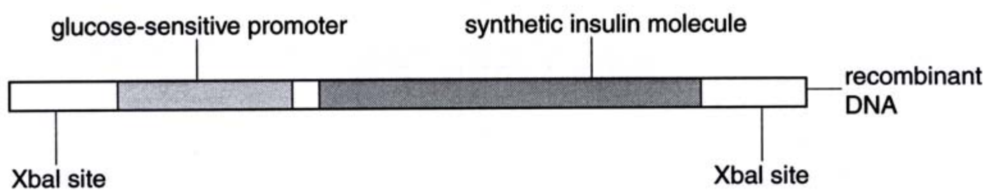
(d) Briefly describe how structure F differs from structure E. [2]

1 **Structure F = ribosome; Structure E = RNA polymerase;**
2 Ribosome composed of **ribosomal proteins and rRNA while polymerase is made up of amino acids;**
3 RNA polymerase is **composed of more than 2 subunits while ribosome consists of 2 subunits (1 large and 1 small subunits);**
4 ribosome is **20 – 30nm; RNA polymerase is smaller in molecular size than ribosomes**;
5 **3 binding sites for ribosomes in E (APE); 1 active site present on RNA polymerase in F;**

[Total: 12]

4 Using genetic engineered bacteria, a synthetic insulin molecule has been produced which has over 20% of the activity of naturally produced insulin. The gene for the synthetic molecule has been used in an experimental gene therapy for insulin – dependent diabetes.

DNA coding for the synthetic molecule was combined with a promoter region (binding site for RNA polymerase during transcription) taken from a gene for an enzyme found only in liver cells. The promoter is sensitive to blood glucose concentration and the synthetic insulin gene will become transcribed when blood glucose concentration rises above normal concentration.



(a) Describe the role of restriction enzyme, like XbaI in genetic engineering. [4]

1 **Cut DNA/ plasmid/ chromosome/ donor gene;**
2 At **specific restriction site** with **palindromic sequences**; **produce sticky ends;**
3 Some RE **produce blunt ends**: add **peptidyl transferase** to change blunt ends to sticky ends;
4 RE is used to cut **donor gene** and **vector** to **create complementary sticky ends;**
5 **Allow donor gene and vector to combine and anneal together based on;**
6 **Complementary base-pairing;** using **DNA ligase; a recombinant DNA is formed;**

(b) Describe how a bacterium that takes up a genetically modified plasmid can be identified. [4]

1 **Antibiotic screening** using **replica plating method;**
2 Plasmid with **two antibiotic resistance genes** are used, and **GoI is inserted into one.**
3 Bacteria **grown in medium containing antibiotic** for which transformed **bacteria have resistance.**
4 Replica **plating done on medium** with **antibiotic that will kill transformed bacteria.**
5 Colonies that survive in the **first antibiotic but not second** = **bacteria that have taken up modified plasmid (inactivation of antibiotic resistance)**
6 OR Using **X-gal reagent** in medium; **insertional inactivation of b – galactosidase gene**
7 **Failure to produce enzyme**: formation of white bacterial colonies = **bacterial colonies that have taken up the modified plasmid;**

Insulin is also synthesized in large scale fermenters from genetically modified bacteria.

(c) Explain advantages to a patient of using insulin produced by genetically modified bacteria. [3]

1 Bovine/Porcine insulin is **different from human insulin → some people may be allergic**,
2 but insulin by bacteria is **synthesized from a human gene**, **hence identical to human insulin.**
3 Also insulin **made by bacteria is pure → synthesized in sterile fermenters.**
4 **Few/no side effects due to similarity with body's own insulin**.

(d) Insulin is a peptide hormone consisting of 51 amino acids. Describe the size of the coding region of the synthetic molecule in the diagram above. Show your working. [2]

Coding region = 51 x 3 (1 mark) = 153 bases (1 mark)
Do not include STOP codon as stop codon is not coding. If stop codon included, 1 mark max.

(e) Suggest an alternative organism that can be used to produce human insulin by genetic transformation.
Give a reason for your answer. [2]

1 **Yeast**
2 **Reason: Yeast is unicellular, asexual reproduction, and is eukaryote, hence perform splicing**.

[Total: 15]

5 In 1882, T. W. Englemann investigated the effect of different wavelengths of light on photosynthesis. He placed a filamentous green alga into a test-tube along with a suspension of motile bacteria which move to regions of high oxygen concentration. He allowed the bacteria to use up the available oxygen and then illuminated the alga with light that had been passed through a prism to form a spectrum. After a short time, he observed the results shown in Fig. 5.1. Bacteria, which are indicated by the tiny rectangles, were distributed evenly throughout the test-tube at the beginning of the experiment.

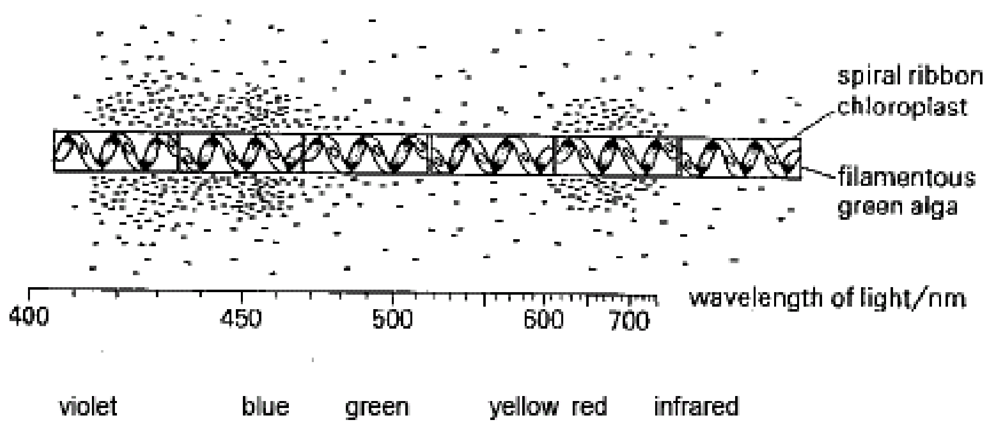
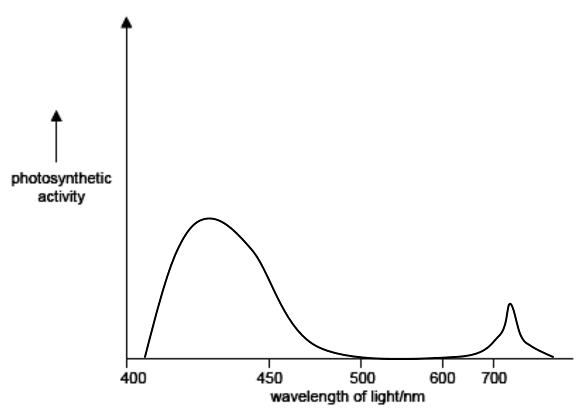


Fig. 4.1

5 **(a)** Using the information provided above, describe how photosynthesis varies with the wavelength of light. [2]

1 Photo synthesis is **maximum at violet – blue (400-475nm) and at 600-700nm (red).**
2 Shows **maximum bacterial growth** at these wavelengths and **almost zero growth** at 500-600nm.



(b) With reference to light dependent reactions, explain the results observed in Fig. 5.1. [4]

1 **photons of light are absorbed by chl a / rxn centre of PSI (P700) & PSII (P680);**
2 pigments with maximal absorption at 680 nm and 700 nm; and also absorption at blue region (400 – 500 nm)
3 accessory pigments like chl b, careteniods, xanthophylls etc. has maximal absorption at blue (400 – 500 nm) and red (600 – 700 nm) regions;
4 allows **photoactivation of PSI**, e- transfer along series of e- carriers to **reduce $NADP^+$ to NADPH;**
5 **photoactivation of PSII**, e- transfer along series of cytochrome carriers to fill e- hole in PSI;
6 **photolysis of water by O_2-evolving complex**, **donate e^-s to PSII, returning it to ground state;** results in formation of O_2 and H^+

(c) State and explain the difference in observations if water supply to the algae was limited. [3]

1 **absence of primary e^- donor**, means e^- in **PSI not replaced;**
2 **$NADP^+$ not reduced**, e^- passed back to **cytochrome then recycled to PSI;**
3 **undergoes cyclic photophosrylation**; **no photolysis of water**, **no O_2 evolution;**
4 bacteria **evenly distributed throughout algae**;

Photosynthesis in the green alga occurs in two stages, at the thylakoid membrane (light dependent) and the stroma (light independent reactions.)

(d) Describe the function of the thylakoid membrane in photophosphorylation. [3]

1 Barrier
Thylakoid membrane **impermeable to H+ ions;** maintains **H+ gradient** from **thylakoid space to stroma;** allows **phosphorylation of ADP to ATP** as **H+ diffuses through ATP synthase**;
2 Organization and localization of function/ Compartmentalisation of function
Attachment of Photosystems and chlorophyll pigments for efficient light absorption; **Increases proximity between electron carrier molecules and photosystems** for light reactions to occur; **Site of attachment of ATP synthase**: ATP synthesizing machinery;
3 **Increases surface area for light absorption;**

(e) Describe the differences in photophosphorylation and oxidative phosphorylation in chloroplast and mitochondria. [5]

1 in **Photophosphorylation(PP),** In the Chloroplast **the reservoir for H+ ion is the Thylakoid membrane**
2 whereas for **Oxidative Phosphorylation (OP**) in the mitochondria the **reservoir is that of the intermembrane space of the mitochondria.**
3 In PP In the chloroplast - electrons are derived from **oxidation of Photosystems (I&II) and the splitting of water in Photolysis.**
4 whereas in the mitochondria, **OP 's electron is initially derived from investments in NADH and FADH** derived **originally from Oxidation of Glucose via Glycolysis and Crebs cycle**.
5 In chloroplasts **ATP synthase synthesizes ATP in the stroma** as protons **diffuse from thylakoid to the stroma**, in mitochondria, **ATP is synthesized in the matrix** as **proton diffuse from intermembrane space to matrix**.

[Total: 17]

6 In the garden pea, *Pisum sativum* two pairs of alleles determine the seed characters green and yellow and round and wrinkled. Table 6.1 shows the results which were obtained from four separate crosses.

Cross	Parent	Progeny			
		Yellow round	Yellow wrinkled	Green round	Green wrinkled
1	Yellow round x green wrinkled	138	143	137	142
2	Yellow round x yellow round	176	0	60	0
3	Yellow round x yellow wrinkled	223	247	75	86
4	Green round x yellow wrinkled	371	0	0	0

(a) State which alleles are dominant and explain your answer. [2]

1 **Yellow is dominant** over green as from **cross 4, all offspring from yellow and green parents are yellow.**

2 **Round** is **dominant** over wrinkled **as from cross 4, all offspring from round and wrinkled seeds are round.**

(b) Using suitable symbols, write down the genotypes of the parents of each cross in the table below. [8]

Round: R, wrinkled: r, Yellow: Y, green: y.

Parents of Cross 1: **RrYy** and **rryy.** P1 P2
Gametes**:** RY, rY, Ry and ry ry
F1: RrYy, rrYy, Rryy, rryy
Phenotype: 1 Round yellow: 1 wrinkled yellow: 1 round green: 1 wrinkled green

Parents of cross 2: **RRYy** and **RRYy** **P1** **P2**
Gametes: RY, Ry RY, Ry
F1: 1 RRYY, 2 RRYy, 1 RRyy
Phenotype: 3 Round yellow: 1 round green.

Parents of cross 3**: RrYy** and **Rryy** P1 P2
Gametes: RY, Ry, rY, ry Ry, ry
F1: RRYy, RRyy, 2RrYy, 2 Rryy , rrYy, rryy
Phenotype: 3 round yellow: 3 round green: 1 wrinkled yellow: 1 wrinkled green

Parents of cross 4: **rrYY** and **RRyy**. P1 P2
Gametes: rY Ry
F1: RrYy
Phenotype: all round yellow.
(1 mark for each correct parental genotype, total 8 marks)

Another experiment was carried out where randomly selected round, yellow plants and green wrinkled plants were self fertilized, and it was found out that only green, wrinkled plants always produced pure-breeding offspring.

(c) Explain why wrinkled, green plants always produce pure breeding offspring while round yellow plants do not. [3]

1 **Wrinkled and green** are **recessive alleles.**
2 Hence they are **expressed only in homozygous condition** and **never in heterozygous**.
3 Hence a plant with **homozygous alleles** will **only produce one kind of gamete (ry)**
4 When **crossed with a gamete from a plant with the same genotype** will produce the **same recessive phenotype.**
5. But for a plant to be yellow and round, **there are four possible genotypes (RrYy, RRYy, RrYY and RRYY)**, all of **which can mask the recessive alleles** but **produce gametes carrying them**.
6 When self **crossed there is possibility of different gametes combining → producing different kinds of offspring**, including those with recessive traits/heterozygous condition.

[Total: 13]

Section B (Answer any ONE question)

7 **(a)** Describe how prokaryotic genome structure differs from eukaryotes, and explain its significance in gene cloning. [10]

Differences:

1 Prokaryotic genome is **circular** while eukaryotic **genome is linear.**
2 Prokaryotic genes are **grouped into operons** with one promoter controlling **expression of many genes** while eukaryotic genes **are not grouped into operons**, each **gene has its own promoter**.
3 Prokaryotic DNA is **free floating in the cytoplasm** as a nucleoid. Eukaryotic genome exists in the **nucleus as distinct chromosomes.**
4 Prokaryotic DNA is made up mostly **of coding regions**, with **few repetitive/satellite DNA**. Only **3% of eukaryotic DNA is coding. Most of it is satellite DNA/repetitive DNA.**
5. Prokaryotic DNA is **not coiled** round **histone proteins**, only **supercoiled. Eukaryotic DNA is coiled around histone proteins and further folded and coiled around scaffold** proteins.
6. Prokaryotic genes have **no introns** and hence, do not have **splicing machinery (spliceosome),** while eukaryotic genes **have introns** and hence **need splicing machinery** to cleave them out and join exons together.
7 Prokaryotic cells have **small circular** pieces of DNA called **plasmids** which are absent in eukaryotes.
8 **Prokaryotic DNA do not need telomeres**, while eukaryotic DNA has **repeating units called telomeres** at the end of the chromosome.

Significance in cloning:

9 Eukaryotic genes to be **cloned into bacteria must be from cDNA,** which is **reverse transcribed from mRNA** because bacteria cannot cleave out introns.
10 Bacteria have plasmids and this allows for **easier transformation** as plasmids can be **manipulated outside cells and inserted into bacteria**.
11 Bacterial genes are grouped into operons which can be inducible or repressible, hence eukaryotic genes must be inserted next to the relevant operon or it would not be expressed.

(b) Using named examples, explain the properties of stem cells. [6]

Property 1

1 Stem cells are **unspecialized/totipotent**
Example of the property using named examples of normal functions

2 An example of such is the **totipotent fertilised egg** and the **first 4 cells** of **embryonic stem cells** produced by its cleavage retains the complete set of genetic information and **have the potential to become any cell type** in the adult body or **any cell** of the **extraembryonic membranes**

Property 2

3 Stem cells are capable of **continually renewing and dividing through cell division** for **long periods**

Example of the property using named examples of normal functions

4 An example of such is **pluripotent stem cells** of the **inner cell mass** of the **blastocyst** that retain the **complete set of genetic information** and **divide continuously** for 9 months **in the formation of a foetus**.

Property 3

5 Stem cells are capable of **differentiating into specialized cell types** under **appropriate conditions**.

Example of the property using named examples of normal functions

6 An example are the **multipotent hematopoietic stem cells** of **adult stem cells** which **give rise to all the types of blood cells**: erythrocytes, B lymphocytes, T lymphocytes, neutrophils, basophils, eosinophils, monocytes, macrophages, and platelets **when there is a local infection** or **injury**.

(c) Discuss the ethical issues involved with the production of genetically modified organisms. [4]

1 **Environmental protection** from GMOs**; safeguards of products with GM components may not be adequate;** to **prevent transfer of genes/pollen** from GM organisms to wild-type organisms**;**

2 **Unknown effects of GM products on human health and disease;** effects may only be known after **long period of exposure; allergic reactions** may occur if people unknowingly consume products containing introduced genes**;**

3 **Tampering with nature / "playing God";** by **mixing genes** among species**;** genetic modifications of animals especially resulting in **enhancement;** draws parallels that **human genetic modification for enhancement one day may occur;**

4 Improper human use of animals - **violations of animal rights**; **cruelty to animals;** results in **animal deformity / disease;**

5 Introduction of **animal genes to plants; / porcine genes to other organisms;** may result in **objections from vegetarian; / specific religious groups;** respectively

6 **Accountability of biotechnological and agricultural firms wrt to GMOs; labeling issues;**

8 **(a)** Outline the main features of the Kreb's Cycle. [8]

Steps:

1 Acetyl CoA combines with a **4 carbon compound**, oxaloacetate (OAA) **to form 6C citrate.**
2 Citrate is **oxidized and decarboxylated** to alpha **ketoglutarate**. Oxidising agent is **NAD.**
3 alpha ketoglutarate is converted to **succinate** by **oxidation and decarboxylation**.
4 Thus, Kreb Cycle is responsible for **the complete breakdown of glucose** and the energy released is transferred to **high energy co-enzymes like NAD and FAD.**
5 Succinate is **oxidized to fumarate and then to malate and to OAA.**
6 This produces **one mole of ATP via GTP synthesis.**
7 Also **produces 1 FADH, NADH.**
8 Hence net yield of **KC is 3 NADH, 1 FADH and 1 ATP per cycle.**
9 Glucose is **completely oxidized to CO_2** and the **energy released from oxidation is stored in NADH and FADH.**
10 Products of Kreb cycle enter **ETC** to **generate more ATP.**

Other Points:

11 **KC occurs in the matrix of the mitochondrion.**
12 KC will only occur in **the presence of oxygen** as in **the absence of O_2 pyruvate will not be converted to acetyl coA to begin the cycle.**

(b) Describe the role of NAD in aerobic respiration. [6]

1 NAD = **co-enzyme**
2 **from intermediates in glycolysis**, transition stage and krebs cycle (= aerobic respiration) **NAD are H^+ and electron carriers which When reduced to NADH.**
3 **Carry the protons and electrons to the ETC for oxidative phosphorylation**
4 Role of NAD in **Glycolysis as oxidizing agent to convert GALP to G3P.**
5 Role of NAD in **Link reaction to convert pyruvate to acetyl coA.**
6 Role of NAD in **Krebs cycle as oxidizing agent to oxidize various intermediates**.
7 Role of **oxidative phosphorylation**, where NADH **donates proton and e^- to electron carriers for the synthesis of ATP.**
8 For the synthesis **of 30 out of the 38 ATP** produced during the complete oxidation of 1 glucose **molecule from 10 molecules of NADH produced in glycolysis**, link reaction & krebs cycle
9 Reoxidized NAD must return to **glycolysis,** recycled to **balance redox.**

(c) Explain the small yield of ATP from anaerobic respiration. [6]

1 Glucose is **not completely broken down**, hence a lot of energy is not released but **stored in intermediate molecules** like **lactate in mammals, ethanol in yeast.**
2 NADH is used to **balance redox** and **regenerate NAD** instead of **going to the ETC.**
3 **In absence of oxygen, pyruvate** is unable to enter mitochondria, **hence remains in cytoplasm**, **cannot undergo oxidative phosphorylation.**
4 Continuation of glycolysis results **in pyruvate accumulation** and **depletion of NADH**, **this stops further synthesis** of ATP **till pyruvate is reduced** and energy is released to accumulate it.
5 **ATP in glycolysis** comes from **only two steps, by substrate level phosphorylation**.
6 **Conversion of G3P to PEP and then to pyruvate, and at each step only 1 ATP is released.**
7 **There is a loss of two ATP right at the start** of glycolysis in the priming reaction, **hence net yield is only 4-2 = 2 ATP.**